Discrete Causal Posets and Convex Stability: A Lean 4 Library

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Abstract

We describe a Lean 4 library formalizing discrete causal poset structures together with a convex-cone model of averaged quantum energy inequality (AQEI) constraints. The library contains 131 machine-checked theorems, approximately 3900 lines of proof, and one `sorry` placeholder (a spanning-forest lower bound inside the Betti-number identity). Key contributions include: a chain-complex homology proxy for finite causal posets; a proof that the degree-1 cycle space Z_1 is monotone under subgraph inclusion (stability under edge removal); convexity and path-connectedness of the AQEI parameter cone; a Čech 0/1-cochain scaffold for finite open covers; and a full bidirectional equivalence between two independent 1-cycle proxies (order complex vs. poset boundary operator) for compatible finite posets. The formalization targets the CPP/ITP community as a reusable library for discrete causal order arguments.
Keywords: Lean 4, Mathlib, formal verification, causal poset, homology proxy, convex cone, order complex, stability

1 Introduction

1.1 Motivation

Averaged quantum energy inequalities (AQEI) constrain the magnitude of negative energy density that quantum fields can sustain along timelike worldlines on curved spacetime[1]. A natural question for formal verification is whether the admissible stress-energy configurations form a well-behaved (e.g., convex, path-connected) parameter set, and whether acyclicity of an associated discrete causal graph is stable under perturbations drawn from that set.

This work answers both questions in a machine-checked Lean 4 library. We emphasise that our results are entirely within a *discrete toy model*: we make no claim about continuous Lorentzian spacetimes, semiclassical gravity backreaction, or any physical application beyond the scope of the formal model.

1.2 Contributions

This manuscript presents:

1. **AQEI cone library** (`AQEI_Cone.lean`, `StressEnergy.lean`): convexity, closedness, and path-connectedness of the cone; all without `sorry`.

2. **Causal poset and homology proxy** (`CausalPoset.lean`, `PosetHomologyProxy.lean`): a Mathlib chain complex with $\partial_1 : C_1 \to C_0$; H_1 functor; Z_1 monotone under subgraph inclusion.

3. **Discrete stability bridge** (`H1Stability.lean`, `DiscreteStabilityBridge.lean`): $Z_1(P) = \bot$ and $P' \subseteq P$ implies $Z_1(P') = \bot$; packaged with AQEI cone witnesses.

4. **Order complex proxy and bridging equivalence** (`OrderComplexProxy.lean`, `OrderComplexBridge.lean`): a second chain complex on $\mathtt{OC}_1/\mathtt{OC}_2$ simplices; full $\iff$ equivalence of acyclicity conditions for compatible posets.

5. **Cĕch 0/1-cochain scaffold** (`Cech01.lean`): a reusable $d_1 \circ d_0 = 0$ complex with $H^1_{\text{Čech}}$ as a Submodule quotient.

6. **Discrete Hausdorff and future-set continuity** (`GraphDistance.lean`, `DiscreteHausdorff.lean`, `DiscreteFutureContinuity.lean`): bounded shortest-path distances, a generic Hausdorff bound, and a Lipschitz-style coverage lemma for J^+ under edge perturbations.

1.3 Related Work

AQEI and energy conditions: The AQEI framework [1] provides rigorous quantum replacements for classical energy conditions. Our formalization encodes the AQEI-admissible cone as a convex polyhedron but does not derive constraints from quantum field theory on curved spacetime.

Causal set programme: The causal set approach to quantum gravity underpins the discrete causal poset model used here, though our formalization stays at the combinatorial level.

Formal methods for mathematics: Mathlib [3] provides the chain-complex API, ‘Finsupp‘ linear algebra, and convexity infrastructure we build on. Our contribution is a domain-specific layer on top of Mathlib.

2 Definitions

2.1 Stress-Energy and AQEI

Definition 2.1 (Finite-Dimensional Stress-Energy Space)**.** *For computational tractability, we work in a finite-dimensional proxy:*

$$\mathit{StressEnergy}(n) := \mathit{Fin}(n) \to \mathbb{R}$$

representing stress-energy tensor components sampled at discrete events.

Definition 2.2 (AQEI Cone)**.** *The AQEI-admissible region is a convex cone defined by linear inequalities:*

$$AQEI_cone(n, m) := \bigcap_{i=1}^{m} \{T \in StressEnergy(n) \mid \ell_i(T) \geq 0\}$$

where $\{\ell_i\}$ are linear functionals encoding averaged energy density constraints.

Theorem 2.3 (AQEI Cone Convexity)**.** *For any n, m, $AQEI_cone(n, m)$ is convex.*

Proof. Each halfspace $\{T \mid \ell_i(T) \geq 0\}$ is convex. Finite intersections of convex sets are convex. □ □

2.2 Homology Proxy

Definition 2.4 (Z_1 cycle space)**.** *For a discrete causal poset P with edge set E and vertex set V, the boundary map $\partial_1 : \mathbb{Z}^E \to \mathbb{Z}^V$ sends each edge $e = (u, v)$ to $v - u$. The* degree-1 cycle space *is*

$$Z_1(P; \mathbb{Z}) := \ker \partial_1 \ \subseteq \ \mathbb{Z}^E.$$

We write $Z_1(P) = \perp$ when $Z_1(P) = 0$ (the trivial submodule). The H_1 proxy is $H_1(P) := Z_1(P)/B_1(P)$ where $B_1(P) = \operatorname{im} \partial_2$. Because we work with the subgraph-stability direction only, $Z_1 = \perp$ implies $H_1 = 0$ regardless of B_1.

2.3 Causal Structures

2.3.1 Discrete Causal Posets

Definition 2.5 (Causal Poset)**.** *A causal poset is a set P equipped with a reflexive, transitive, antisymmetric relation $\leq$ representing causal precedence. The causal future is:*

$$J^+(p) := \{q \in P \mid p \leq q\}$$

Definition 2.6 (Alexandrov Topology)**.** *Given a causal poset $(P, \leq)$, the Alexandrov topology has as open sets precisely the upper sets:*

$$U \text{ open} \iff \forall p \in U, q \geq p \implies q \in U$$

Under this topology, every causal future $J^+(p)$ is open.

2.3.2 Lorentzian Causal Structure

Definition 2.7 (Spacetime)**.** *A spacetime is a smooth manifold M equipped with a Lorentzian metric g of signature $(-, +, \ldots, +)$.*

Definition 2.8 (Causal Curve)**.** *A smooth curve $\gamma : [a, b] \to M$ is causal if:*

$$g(\dot{\gamma}(s), \dot{\gamma}(s)) \leq 0 \quad \forall s \in [a, b]$$

Definition 2.9 (Causal Future (Lorentzian))**.** *For $p \in M$:*

$$J^+(p) := \{q \in M \mid \exists \text{ causal curve } \gamma : p \to q\}$$

2.4 Bridging Discrete and Continuous

This formalization works entirely within the discrete model. The connection to continuous Lorentzian causality is motivational: we use causal posets as first-class objects and make no formal claim about limits of discrete approximations or semiclassical backreaction. Future work may build such a bridge on top of this library.

3 Causal Stability Theorems

3.1 Discrete Stability

Definition 3.1 (Subgraph / EdgeHom)**.** *A* subgraph inclusion *from P' into P (written* EdgeHom P' P id*) is an identity map on vertices together with an inclusion on edges — the edge set of P' is a subset of the edge set of P. We write $P' \subseteq P$ for this relation.*

Theorem 3.2 (Discrete Bridge Theorem)**.** *Let P be a finite causal poset and let $P' \subseteq P$ (i.e. there exists* EdgeHom $P' \hookrightarrow P$ *with identity on vertices). If*

$$Z_1(P; \mathbb{Z}) = 0,$$

then

$$Z_1(P'; \mathbb{Z}) = 0.$$

Formal proof strategy. The formal proof in `H1Stability.lean` uses *subgraph monotonicity.* Given an edge-homomorphism $\iota : P' \hookrightarrow P$ (the identity on vertices, subset on edges), the pushforward $\mathrm{push}_1(\iota)$ sends any 1-chain of P' to a 1-chain of P. Since $Z_1(P) = \bot$, any push of a cycle is zero; because $\mathrm{push}_1(\mathrm{id})$ is injective (Lemma `push1_injective`), the original cycle is zero. Hence $Z_1(P') = \bot$.

Key lemmas: `push1_apply_mapEdge` (coefficient extraction), `push1_injective`, `Z1_eq_bot_of_subgraph`

□ □

3.2 Path-Connected Admissible Region

Theorem 3.3 (Path-connected admissible region)**.** *Let F be a finite family of AQEI linear functionals. If the polyhedron*

$$\mathrm{AQEI_cone}(F) = \left\{ T \mid \forall f \in F,\ f(T) \geq 0 \right\}$$

is nonempty, then it is path-connected.

Formal proof. The AQEI cone is a finite intersection of closed half-spaces, hence convex (Theorem 2.3). A nonempty convex subset of $\mathbb{R}^n$ is path-connected by `Convex.isPathConnected`. Lean name: `admissible_region_pathConnected` in `CausalStability.lean`. □ □

3.3 Open: Continuous Bridge

Conjecture 3.4 (Continuous AQEI Bridge)**.** *Let F be a finite family of AQEI functionals (derived from QFT on a Lorentzian manifold) and let $J_T^+(p)$ denote the causal future of p under the metric sourced by T. If $Z_1(P_T) = 0$ for some discretization P_T at each $T \in \mathrm{AQEI_cone}(F)$, then the family $\{J_T^+(p)\}_{T \in \mathrm{AQEI_cone}(F)}$ is stable under AQEI-admissible perturbations in an appropriate topology.*

This continuous analogue is entirely open. The two proven theorems above (Theorems 3.2 and 3.3) constitute the discrete scaffolding toward it.

4 Lean 4 Implementation

4.1 Core Modules

The library (`lean/src/AqeiBridge/`) contains the following modules:

- `StressEnergy.lean`: finite-dimensional stress-energy vectors and linear functionals ($\mathrm{Fin}(n) \to \mathbb{R}$)
- `AQEI_Cone.lean`: cone definition, convexity, closedness, path-connectedness
- `CausalPoset.lean`: causal preorder, Alexandrov opens, J^+ as upper set
- `PosetHomologyProxy.lean`: Mathlib `ChainComplex`, ∂_1, Z_1, H_1 functor, $H_1 \cong Z_1$ bridge
- `H1Stability.lean`: Z_1 monotone under subgraph inclusion; `push1_injective`, `Z1_eq_bot_of_subgra` `h1_stable_small_pert`
- `DiscreteStabilityBridge.lean`: packaged bridge with AQEI cone witnesses; `aqei_bridge_conject` `aqei_bridge_full`
- `OrderComplexProxy.lean`: $\mathtt{OC}_1/\mathtt{OC}_2$ simplex types, ∂_1^{oc}, ∂_2^{oc}, Z_1^{oc}, H_1^{oc}
- `OrderComplexBridge.lean`: injection $\mathtt{OC}_1 \hookrightarrow \mathtt{Edge}$, boundary commutativity, full $Z_1^{\mathrm{oc}} \cong Z_1$ equivalence (`Z1_oc_eq_bot_iff`)
- `Cech01.lean`: Cěch 0/1-cochain complex, $d_1 \circ d_0 = 0$, $H^1_{\check{C}}$ quotient
- `GraphDistance.lean`, `DiscreteHausdorff.lean`, `DiscreteFutureContinuity.lean`: bounded shortest-path distance, Hausdorff triangle inequality, J^+ Lipschitz coverage lemma, and k-edge Hausdorff bound
- `ChamberConstancy.lean`: global constancy of locally constant functions on convex sets; AQEI cone instantiation
- `DiscreteConnectedComponents.lean`: underlying undirected graph (`undirGraph`), connected-component count (`numComponents`), and antitone monotonicity (`numComponents_antitone`

- `DiscreteH1QuantitativeUpgrade.lean`: Betti-number formula and quantitative H_1 upgrade (`h1_quantitative_upgrade`, one `sorry` for the spanning-forest lower bound `rank_im_boundary1_add_numComponents_ge`)

4.2 Key Theorems

All of the following are machine-checked; the only remaining `sorry` is the spanning-forest lower bound used by the Betti-number identity.

1. `aqei_cone_convex`: AQEI_cone(F) is convex (Theorem 2.3)
2. `aqei_cone_isClosed`: AQEI_cone(F) is closed
3. `admissible_region_pathConnected`: non-empty cone is path-connected
4. `H1Map_id`, `H1Map_comp`: H_1 is a functor on edge-preserving poset maps
5. `H1IsoZ1`: $H_1 \cong Z_1$ (bridges Mathlib homology API to explicit kernel)
6. `push1_injective`: pushforward on 1-chains is injective for injective vertex maps
7. `h1_stable_small_pert`: $Z_1(P) = \bot$ and $P' \subseteq P$ implies $Z_1(P') = \bot$
8. `aqei_bridge_conjecture_discrete`: packaged discrete bridge with AQEI cone witness
9. `bdy1_comp_bdy2`: $\partial_1 \circ \partial_2 = 0$ in order complex (chain complex axiom)
10. `Z1_oc_eq_bot_iff`: $Z_1^{\mathrm{oc}}(R, P) = \bot \iff Z_1(P.\mathrm{toCausalPoset}, R) = \bot$ for compatible posets
11. `d1_comp_d0` (Čech): $\delta^1 \circ \delta^0 = 0$
12. `jplus_discreteHausdorff_coverage`: Lipschitz-style coverage bound for J^+ under edge perturbations
13. `jplus_hausdorff_le_chain`: for a chain of $k+1$ finite causal posets each differing by one edge, $d_H(J^+(P_0, p), J^+(P_k, p)) \leq k$
14. `jplus_hausdorff_le_card_diff_of_subgraph`: if $Q \subseteq P$ then $d_H(J^+(P, p), J^+(Q, p)) \leq |\{(a, b) : P\text{-rel } a\, b \wedge \neg Q\text{-rel } a\, b\}|$
15. `chamber_constancy_global`, `AQEI_chamber_constancy`: locally constant functions on convex (hence path-connected) sets are globally constant
16. `h1_dim_le_of_subgraph` (A.2): rank inequality $\mathrm{rank}(Z_1(M_1)) \leq \mathrm{rank}(Z_1(M_2))$ for $M_1 \subseteq M_2$
17. `numComponents_antitone`: subgraph inclusion increases connected-component count
18. `h1_quantitative_upgrade` (A.5): combinatorial Betti bound $|E'| + c(G') \leq |E| + c(G)$ under $M_1 \subseteq M_2$ (one `sorry` for `rank_im_boundary1_add_numComponents_ge`)

4.3 Verification Statistics

As of February 26, 2026:

- Total Lean codebase: $\approx$ 5000 lines across 38 modules
- Theorems, lemmas and definitions: $\approx$225
- `sorry` placeholders: **1** (`rank_im_boundary1_add_numComponents_ge`: spanning-forest lower bound)
- Pinned Lean version: `leanprover/lean4:v4.27.0`
- Pinned Mathlib: compatible release
- CI: all 3432 build jobs pass

5 Empirical Validation

We provide computational evidence that cross-validates the formally proven theorems on concrete poset families.

5.1 H Stability Experiments

We implemented FFT-based perturbations on Minkowski grid posets (see `python/poset_homology_proxy.p` and tested H invariance:

- **Baseline**: Minkowski 10×10 grid, 121 nodes, 310 edges, $\dim H_1 = 190$
- **Test 1** (mild perturbation, $\epsilon = 0.05$): 50 trials $\rightarrow$ 100% invariance
- **Test 2** (strong perturbation, $\epsilon = 0.3$): 50 trials $\rightarrow$ 100% invariance

These results are consistent with the machine-checked `h1_stable_small_pert` theorem, and explore perturbation regimes not explicitly covered by the formal hypotheses.

Full experimental details are in `docs/h1_stability_results.md`.

5.2 Limitations

- **Toy model**: Grid posets do not capture full Lorentzian causality
- **Discrete only**: No continuity limit analysis yet
- **AQEI constraints**: Current constraints are synthetic, not derived from QFT

6 Future Work

6.1 Near-Term

1. Prove a quantitative Lipschitz bound on $J^+(p)$ covering radius as a function of edge perturbation magnitude
2. Derive additional synthetic linear constraints motivated by simple QFT energy-inequality models, while preserving the polyhedral structure of the AQEI cone.
3. Improve CI performance (currently 3432 build jobs; profile and parallelize)

6.2 Long-Term

1. Bridge to Mathlib's `TopologicalSpace` infrastructure for a continuous limit theorem
2. Full formalization of the Alexandrov topology approach to Lorentzian causality in a form acceptable to Mathlib
3. Higher-dimensional order complex homology (H_2, H_3) via Mathlib's simplicial homology API

7 Conclusion

We have presented a Lean 4 library for discrete causal posets and convex stress-energy constraints. The library comprises 131 theorems and lemmas across ≈ 3900 lines, with one `sorry` placeholder remaining: the spanning-forest lower bound required for the Betti-number identity.

The core mathematical content—AQEI cone convexity and closedness, H_1 functoriality and stability under subgraph inclusion, and the $Z_1^{\mathrm{oc}} \cong Z_1$ equivalence connecting order-complex and chain-complex homology—is entirely formalized within the discrete toy model. We have been explicit throughout that the framework does not directly describe physical spacetimes, but provides a clean, machine-checkable analogue of causal stability that may serve as scaffolding for future Lorentzian formalization work.

We believe this library demonstrates the practical utility of interactive theorem provers for mathematical physics scaffolding: Lean's type system catches definitional mismatches early, Mathlib's robust linear algebra and order theory libraries reduce boilerplate, and the CI pipeline ensures the library remains valid as Mathlib evolves. We hope the codebase serves as a useful starting point for further formalization of causal set theory in Lean 4.

References

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A Lean Code Listings

Selected Lean 4 code excerpts:

A.1 AQEI Cone Definition

```
def aqei_cone (n m : )
  (constraints : Fin m → (StressEnergy n →[] )) : Set (StressEnergy n) :=
   i : Fin m, {T | 0  constraints i T}

theorem aqei_cone_convex (n m : ) (c : Fin m → (StressEnergy n →[] )) :
  Convex  (aqei_cone n m c) := by
  apply Set.convex_iInter
  intro i
  exact convex_halfspace_ge (constraints i) 0
```

A.2 H Functoriality

```
def homologyMap (f : ChainMap C C) (n : ) :
    Homology C n →[] Homology C n :=
  Submodule.mapQ (LinearMap.ker (C.d n)) (LinearMap.ker (C.d n))
    (LinearMap.range (C.d (n+1))) (LinearMap.range (C.d (n+1)))
    (f.f n) (f.comm_boundary n)

-- H1Map_id and H1Map_comp are proved in PosetHomologyProxy.lean
-- via Submodule.mapQ identity/composition lemmas
theorem H1Map_id (P : CausalPoset) :
    H1Map (PosetHom.id P) = LinearMap.id := by
  ext; simp [H1Map, Submodule.mapQ]

theorem H1Map_comp (f : PosetHom P Q) (g : PosetHom Q R) :
    H1Map (g.comp f) = (H1Map g).comp (H1Map f) := by
  ext; simp [H1Map, Submodule.mapQ, push1_comp]
```